



# Database Management Systems

## Data Definition SQL queries

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# Database Management Systems

## Unit 1: Introduction to Database Management and SQL

Slides adapted from Author Slides of “Database System Concepts”, Silberschatz, H Korth and S Sudarshan, McGrawHill, 7th Edition, 2019. And Author Slides of Fundamentals of Database Systems”, Ramez Elamsri, Shamkant B Navathe, Pearson, 7th Edition, 2017.

Special Acknowledgement to all CSE Professors who contributed to this endeavor

**Dr. Nagasundari S**

Department of Computer Science and Engineering



- **CREATE** - to create objects in the database
- **ALTER** - alters the structure of the database
- **DROP** - delete objects from the database
- **TRUNCATE** - remove all records from a table, including all spaces allocated for the records are removed
- **COMMENT** - add comments to the data dictionary
- **RENAME** - rename an object

# Database Management Systems

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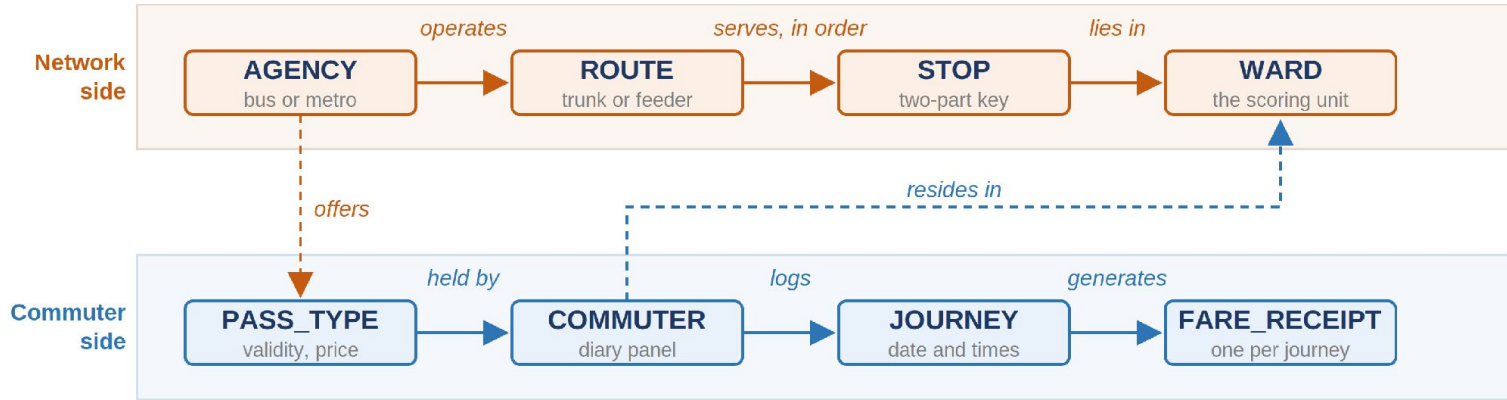
## Unit 1: Introduction to Database Management and SQL

### A Case Study based DDL Queries Execution



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## Transit Gap Finder - the case study in one picture



### Score how well every ward in Bengaluru is served by public transport

More than one operator runs the network, so an agency sits at the top of the chain and a stop code means nothing without it. A route may feed another route. Every stop, and every commuter on the travel-diary panel, belongs to exactly one ward - which is what makes the score computable. A completed journey generates exactly one fare receipt.

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## Six rules the schema has to enforce

### 1 Stop code is local

BMTC and the metro both issue S001, so STOP needs a two-part key

### 2 Platform hangs off STOP

a repeating detail of a dependent table, so its key has three parts

### 3 A stop must have a ward

every stop lies in exactly one, so Ward\_ID is NOT NULL

### 4 One receipt per journey

a foreign key plus UNIQUE allows only one per journey

### 5 A route may feed a route

one nullable column pointing back at the same table

### 6 Age is never stored

computed values live in a view, or the stored copy goes stale

## Each rule turns into one specific piece of DDL in the statements that follow

A composite primary key, a NOT NULL column, a UNIQUE on a foreign key, a nullable self-reference, and two views. Nothing is stored twice, nothing computed is stored at all.

### Transit Gap Finder: BMTC + Metro Accessibility Scoring

Score how well every ward in Bengaluru is served by public transport.

### The twelve relations this demo creates

AGENCY, WARD, ROUTE, PASS\_TYPE, COMMUTER, JOURNEY, FARE\_RECEIPT

STOP - dependent on AGENCY, primary key (Agency\_ID, Stop\_Code)

PLATFORM - repeating detail of STOP, three-part key

SERVES, HOLDS - link tables; COMMUTER\_PHONE - repeating column

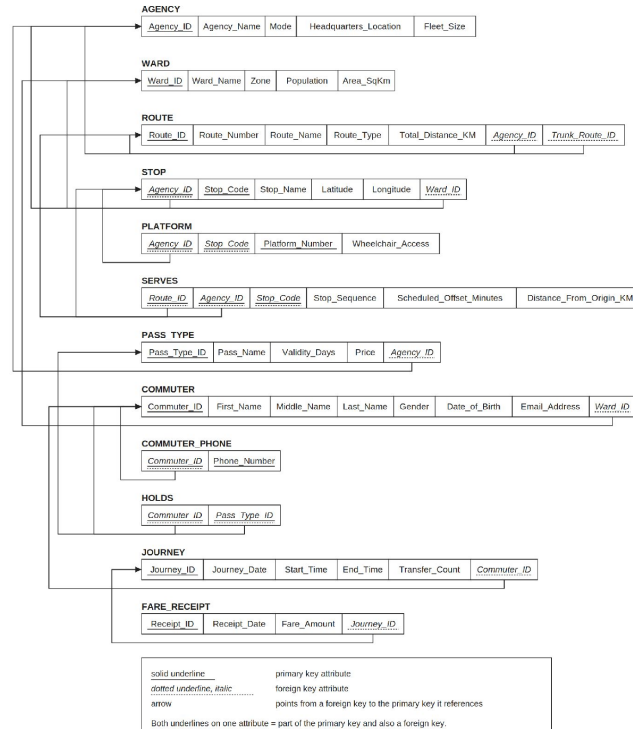
### What to watch for while the commands run

1. STOP needs a composite key AND a mandatory Ward\_ID
2. Trunk\_Route\_ID is a self-referencing foreign key - never CASCADE
3. Age and Population\_Density are computed, so they live in views

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## The relational schema - twelve relations

Arrows run from each foreign key to the primary key it references.





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## Demo setup - prompt and database



### Set the prompt so every screenshot is self-identifying

```
prompt (SRN@\U) [\d] > \_
```

### Create and select the case study database

```
CREATE DATABASE IF NOT EXISTS transit_gap;  
USE transit_gap;  
SHOW DATABASES;
```

```
MySQL 8.0 Command Line C# x + v  
Enter password: *****  
Welcome to the MySQL monitor.  Commands end with ; or \g.  
Your MySQL connection id is 9  
Server version: 8.0.46 MySQL Community Server - GPL  
  
Copyright (c) 2000, 2026, Oracle and/or its affiliates.  
  
Oracle is a registered trademark of Oracle Corporation and/or its  
affiliates. Other names may be trademarks of their respective  
owners.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
mysql> -- =====  
mysql> -- Database Management Systems - Department of CSE  
mysql> -- Case study for slides - 03  
mysql> -- ER to Relational Schema Mapping and DDL Query Execution  
mysql> -- Transit Gap Finder: BMTTC + Metro Accessibility Scoring  
mysql> --  
mysql> -- SRN : PES1UG23CS024  
mysql> -- File: PES1UG23CS024_DDL_Lab3.sql  
mysql> --  
mysql> -- Paste one block at a time and screenshot the result.  
mysql> -- Block headings match the slide titles in DDL_Demo.pptx.  
mysql> -- Tested on MySQL 8.0 (InnoDB).  
mysql> -- =====  
mysql> --  
mysql> -- =====  
mysql> -- DEMO SETUP - session prompt and database
```

```
MySQL 8.0 Command Line C# x + v  
mysql> -- =====  
mysql> -- DEMO SETUP - session prompt and database  
mysql> -- =====  
mysql> -- NOTE: "prompt" is a mysql client command, not SQL. It works when you  
mysql> -- paste it at the mysql> prompt. It will NOT work inside SOURCE.  
mysql> --  
mysql> prompt (PES1UG23CS024@U) [\d] > \_  
PROMPT set to '(PES1UG23CS024@U) [\d] > \_'  
(PES1UG23CS024@root@localhost) [(none)] >  
(PES1UG23CS024@root@localhost) [(none)] > CREATE DATABASE IF NOT EXISTS transit_gap;  
Query OK, 1 row affected (0.01 sec)  
  
(PES1UG23CS024@root@localhost) [(none)] > USE transit_gap;  
Database changed  
(PES1UG23CS024@root@localhost) [transit_gap] > SHOW DATABASES;  
+-----+  
| Database |  
+-----+  
| information_schema |  
| movies |  
| mysql |  
| performance_schema |  
| sakila |  
| smart_hostel |  
| sys |  
| transit_gap |  
| world |  
+-----+  
9 rows in set (0.02 sec)
```

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## 2.1a CREATE TABLE - the two parent tables



```
CREATE TABLE AGENCY (  
    Agency_ID          VARCHAR(10)  NOT NULL,  
    Agency_Name        VARCHAR(60)  NOT NULL,  
    Mode               VARCHAR(10)  NOT NULL DEFAULT 'BUS',  
    Headquarters_Location VARCHAR(100) NOT NULL,  
    Fleet_Size         INT           NOT NULL DEFAULT 0,  
    CONSTRAINT pk_agency PRIMARY KEY (Agency_ID),  
    CONSTRAINT uq_agency_name UNIQUE (Agency_Name),  
    CONSTRAINT chk_agency_mode CHECK (Mode IN ('BUS', 'METRO')));
```

```
CREATE TABLE WARD (  
    Ward_ID   VARCHAR(10)  NOT NULL,  
    Ward_Name VARCHAR(60)  NOT NULL,  
    Area_SqKm DECIMAL(8,3) NOT NULL,  
    CONSTRAINT pk_ward PRIMARY KEY (Ward_ID));
```

A screenshot of a MySQL command-line interface window. The window title is 'MySQL 5.6 Command Line CS'. It shows the execution of two SQL scripts. The first script creates the 'AGENCY' table with columns Agency\_ID, Agency\_Name, Mode, Headquarters\_Location, and Fleet\_Size, and includes primary, unique, and check constraints. The second script creates the 'WARD' table with columns Ward\_ID, Ward\_Name, Area\_SqKm, and includes primary, unique, and check constraints. Both scripts are executed successfully, with messages indicating 0 rows affected and execution times of 0.03 seconds and 0.01 seconds respectively.

```
MySQL 5.6 Command Line CS x -  
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====  
(PES1UG23CS024@root@localhost) [transit_gap] > -- 2.1a CREATE TABLE - the two parent tables  
(PES1UG23CS024@root@localhost) [transit_gap] > =====  
(PES1UG23CS024@root@localhost) [transit_gap] > CREATE TABLE AGENCY (  
-> Agency_ID          VARCHAR(10)  NOT NULL,  
-> Agency_Name        VARCHAR(60)  NOT NULL,  
-> Mode               VARCHAR(10)  NOT NULL DEFAULT 'BUS',  
-> Headquarters_Location VARCHAR(100) NOT NULL,  
-> Fleet_Size         INT           NOT NULL DEFAULT 0,  
-> CONSTRAINT pk_agency PRIMARY KEY (Agency_ID),  
-> CONSTRAINT uq_agency_name UNIQUE (Agency_Name),  
-> CONSTRAINT chk_agency_mode CHECK (Mode IN ('BUS', 'METRO')),  
-> CONSTRAINT chk_agency_fleet CHECK (Fleet_Size >= 0)  
-> );  
Query OK, 0 rows affected (0.03 sec)  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > CREATE TABLE WARD (  
-> Ward_ID   VARCHAR(10)  NOT NULL,  
-> Ward_Name VARCHAR(60)  NOT NULL,  
-> Area_SqKm DECIMAL(8,3) NOT NULL,  
-> CONSTRAINT pk_ward PRIMARY KEY (Ward_ID),  
-> CONSTRAINT uq_ward_name UNIQUE (Ward_Name),  
-> CONSTRAINT chk_ward_area CHECK (Area_SqKm >= 0)  
-> );  
Query OK, 0 rows affected (0.01 sec)
```

# Database Management Systems

## 2.1b CREATE TABLE - STOP and its two-part key



**STOP is identified by AGENCY, so its key is two-part**

```
CREATE TABLE STOP (  
    Agency_ID VARCHAR(10) NOT NULL,  
    Stop_Code VARCHAR(12) NOT NULL,  
    Stop_Name VARCHAR(80) NOT NULL,  
    Ward_ID VARCHAR(10) NOT NULL,  
    CONSTRAINT pk_stop PRIMARY KEY (Agency_ID, Stop_Code));
```

**PLATFORM repeats within one STOP - three-part key**

```
CREATE TABLE PLATFORM (  
    Agency_ID VARCHAR(10) NOT NULL,  
    Stop_Code VARCHAR(12) NOT NULL,  
    Platform_Number VARCHAR(5) NOT NULL,  
    CONSTRAINT pk_platform PRIMARY KEY  
        (Agency_ID, Stop_Code, Platform_Number));
```

```
MSQL 8.0 Command Line Client  
[PES1UG23CS824@rootlocalhost] [transit_gap] > -----  
[PES1UG23CS824@rootlocalhost] [transit_gap] > -- 2.1b CREATE TABLE - the weak entity STOP and its multivalued attribute  
[PES1UG23CS824@rootlocalhost] [transit_gap] > -----  
[PES1UG23CS824@rootlocalhost] [transit_gap] > -- STOP is identified by AGENCY, so its primary key is two-part.  
[PES1UG23CS824@rootlocalhost] [transit_gap] > -- Ward_ID is NOT NULL because STOP participates totally in LIES.IN.  
[PES1UG23CS824@rootlocalhost] [transit_gap] > CREATE TABLE STOP (  
-> Agency_ID VARCHAR(10) NOT NULL,  
-> Stop_Code VARCHAR(12) NOT NULL,  
-> Stop_Name VARCHAR(80) NOT NULL,  
-> Latitude DECIMAL(9,6) NOT NULL,  
-> Longitude DECIMAL(9,6) NOT NULL,  
-> Ward_ID VARCHAR(10) NOT NULL,  
-> CONSTRAINT pk_stop PRIMARY KEY (Agency_ID, Stop_Code),  
-> CONSTRAINT ch_stop_lon CHECK (Latitude BETWEEN -90 AND 90),  
-> CONSTRAINT ch_stop_lon CHECK (Longitude BETWEEN -180 AND 180)  
-> );  
Query OK, 0 rows affected (0.01 sec)  
[PES1UG23CS824@rootlocalhost] [transit_gap] > -- PLATFORM needs a THREE-part key: the owner STOP already has two.  
[PES1UG23CS824@rootlocalhost] [transit_gap] > CREATE TABLE PLATFORM (  
-> Agency_ID VARCHAR(10) NOT NULL,  
-> Stop_Code VARCHAR(12) NOT NULL,  
-> Platform_Number VARCHAR(5) NOT NULL,  
-> Wheelchair_Access CHAR(1) NOT NULL DEFAULT 'N',  
-> CONSTRAINT pk_platform PRIMARY KEY (Agency_ID, Stop_Code, Platform_Number),  
-> CONSTRAINT ch_platform_access CHECK (Wheelchair_Access IN ('Y','N'))  
-> );
```

```
MSQL 8.0 Command Line Client  
[PES1UG23CS824@rootlocalhost] [transit_gap] > -- PLATFORM needs a THREE-part key: the owner STOP already has two.  
[PES1UG23CS824@rootlocalhost] [transit_gap] > CREATE TABLE PLATFORM (  
-> Agency_ID VARCHAR(10) NOT NULL,  
-> Stop_Code VARCHAR(12) NOT NULL,  
-> Platform_Number VARCHAR(5) NOT NULL,  
-> Wheelchair_Access CHAR(1) NOT NULL DEFAULT 'N',  
-> CONSTRAINT pk_platform PRIMARY KEY (Agency_ID, Stop_Code, Platform_Number),  
-> CONSTRAINT ch_platform_access CHECK (Wheelchair_Access IN ('Y','N'))  
-> );  
Query OK, 0 rows affected (0.01 sec)  
[PES1UG23CS824@rootlocalhost] [transit_gap] > -----  
[PES1UG23CS824@rootlocalhost] [transit_gap] > -- 2.1c CREATE TABLE - ROUTE and the M:N relation SERVES  
[PES1UG23CS824@rootlocalhost] [transit_gap] > -----  
[PES1UG23CS824@rootlocalhost] [transit_gap] > -- Trunk_Route_ID is the recursive FEEDS relationship. It MUST allow NULL  
[PES1UG23CS824@rootlocalhost] [transit_gap] > CREATE TABLE ROUTE (  
-> Route_ID VARCHAR(10) NOT NULL,  
-> Route_Name VARCHAR(80) NOT NULL,  
-> Route_Type VARCHAR(20) NOT NULL DEFAULT 'ORDINARY',  
-> Total_Distance_M DECIMAL(6,2) NOT NULL,  
-> Agency_ID VARCHAR(10) NOT NULL,  
-> Trunk_Route_ID VARCHAR(10) NULL,  
-> CONSTRAINT pk_route PRIMARY KEY (Route_ID),  
-> CONSTRAINT ch_route_distance CHECK (Total_Distance_M > 0)
```

```
CREATE TABLE ROUTE (  
    Route_ID          VARCHAR(10) NOT NULL,  
    Route_Number      VARCHAR(15) NOT NULL,  
    Route_Type        VARCHAR(20) NOT NULL DEFAULT 'ORDINARY',  
    Agency_ID         VARCHAR(10) NOT NULL,  
    Trunk_Route_ID    VARCHAR(10) NULL,          -- self-reference  
    CONSTRAINT pk_route PRIMARY KEY (Route_ID));
```

### SERVES carries three columns of its own

```
CREATE TABLE SERVES (  
    Route_ID VARCHAR(10) NOT NULL, Agency_ID VARCHAR(10) NOT NULL,  
    Stop_Code VARCHAR(12) NOT NULL, Stop_Sequence INT NOT NULL,  
    Scheduled_Offset_Minutes INT NOT NULL DEFAULT 0,  
    Distance_From_Origin_KM DECIMAL(6,2) NOT NULL DEFAULT 0.00,  
    CONSTRAINT pk_serves PRIMARY KEY (Route_ID, Agency_ID, Stop_Code));
```

```
MySQL 8.0 Command Line Client > -- =====  
(PES1UG23CS024@root@localhost) [transit_gap] > -- 2.1c CREATE TABLE - ROUTE and the M:N relation SERVES  
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====  
(PES1UG23CS024@root@localhost) [transit_gap] > Trunk_Route_ID is the recursive FEEDS relationship. It must allow NULL.  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > CREATE TABLE ROUTE (  
-> Route_ID          VARCHAR(10) NOT NULL,  
-> Route_Number      VARCHAR(15) NOT NULL,  
-> Route_Type        VARCHAR(20) NOT NULL,  
-> Agency_ID         VARCHAR(10) NOT NULL,  
-> Trunk_Route_ID    VARCHAR(10) NULL,  
-> CONSTRAINT pk_route PRIMARY KEY (Route_ID),  
-> CONSTRAINT chk_route_distance CHECK (Total_Distance_KM > 0)  
-> );  
Query OK, 0 rows affected (0.01 sec)
```

```
MySQL 8.0 Command Line Client > -- =====  
(PES1UG23CS024@root@localhost) [transit_gap] > CREATE TABLE SERVES (  
-> Route_ID          VARCHAR(10) NOT NULL,  
-> Agency_ID         VARCHAR(10) NOT NULL,  
-> Stop_Code         VARCHAR(12) NOT NULL,  
-> Stop_Sequence     INT NOT NULL,  
-> Scheduled_Offset_Minutes INT NOT NULL DEFAULT 0,  
-> Distance_From_Origin_KM DECIMAL(6,2) NOT NULL DEFAULT 0.00,  
-> CONSTRAINT pk_serves PRIMARY KEY (Route_ID, Agency_ID, Stop_Code),  
-> CONSTRAINT uq_serves_sequence UNIQUE (Route_ID, Stop_Sequence),  
-> CONSTRAINT chk_serves_sequence CHECK (Stop_Sequence > 0),  
-> CONSTRAINT chk_serves_offset CHECK (Scheduled_Offset_Minutes >= 0)  
-> );  
Query OK, 0 rows affected (0.02 sec)
```

```
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====  
(PES1UG23CS024@root@localhost) [transit_gap] > CREATE TABLE - PASS_TYPE, COMMUTER and its multivalued attribute  
(PES1UG23CS024@root@localhost) [transit_gap] > CREATE TABLE PASS_TYPE (  
-> Pass_Type_ID     VARCHAR(10) NOT NULL,  
-> Pass_Name        VARCHAR(50) NOT NULL,  
-> Validity_Days    INT NOT NULL DEFAULT 1,  
-> Price            DECIMAL(10,2) NOT NULL,  
-> Agency_ID        VARCHAR(10) NOT NULL,  
-> CONSTRAINT pk_pass_type PRIMARY KEY (Pass_Type_ID),  
-> CONSTRAINT chk_pass_validity CHECK (Validity_Days > 0),  
-> CONSTRAINT chk_pass_price CHECK (Price >= 0)
```

```
CREATE TABLE COMMUTER (  
    Commuter_ID    VARCHAR(12) NOT NULL,  
    First_Name     VARCHAR(30) NOT NULL,  
    Last_Name      VARCHAR(30) NOT NULL,  
    Gender         CHAR(1)     NOT NULL,  
    Date_of_Birth  DATE        NOT NULL,  
    Email_Address  VARCHAR(60) NOT NULL,  
    Ward_ID        VARCHAR(10) NOT NULL,  
    CONSTRAINT pk_commuter PRIMARY KEY (Commuter_ID),  
    CONSTRAINT uq_commuter_email UNIQUE (Email_Address));
```

```
CREATE TABLE COMMUTER_PHONE (  
    Commuter_ID    VARCHAR(12) NOT NULL,  
    Phone_Number   VARCHAR(15) NOT NULL,  
    CONSTRAINT pk_commuter_phone PRIMARY KEY (Commuter_ID, Phone_Number));
```

```
MySQL 8.0 Command-Line Client > -- =====  
(PES1UG23C5924@root@localhost) [transit_gap] > -- 2.1d CREATE TABLE ~ PASS_TYPE, COMMUTER and its multivalued attribute  
(PES1UG23C5924@root@localhost) [transit_gap] > -- =====  
(PES1UG23C5924@root@localhost) [transit_gap] > CREATE TABLE PASS_TYPE (  
->    Pass_Type_ID VARCHAR(18) NOT NULL,  
->    Pass_Name    VARCHAR(50) NOT NULL,  
->    Validity_Days INT NOT NULL DEFAULT 1,  
->    Price         DECIMAL(18,2) NOT NULL,  
->    Agency_ID    VARCHAR(18) NOT NULL,  
->    CONSTRAINT pk_pass_type PRIMARY KEY (Pass_Type_ID),  
->    CONSTRAINT ch_pass_validity CHECK (Validity_Days > 0),  
->    CONSTRAINT ch_pass_price CHECK (Price > 0)  
-> );  
Query OK, 0 rows affected (0.01 sec)  
(PES1UG23C5924@root@localhost) [transit_gap] >  
(PES1UG23C5924@root@localhost) [transit_gap] > CREATE TABLE COMMUTER (  
->    Commuter_ID VARCHAR(12) NOT NULL,  
->    First_Name  VARCHAR(30) NOT NULL,  
->    Middle_Name VARCHAR(30) NULL,  
->    Last_Name   VARCHAR(30) NOT NULL,  
->    Gender      CHAR(1) NOT NULL,  
->    Date_of_Birth DATE NOT NULL,  
->    Email_Address VARCHAR(60) NOT NULL,  
->    Ward_ID     VARCHAR(10) NOT NULL,  
->    CONSTRAINT pk_commuter PRIMARY KEY (Commuter_ID),  
->    CONSTRAINT uq_commuter_email UNIQUE (Email_Address)  
-> );
```

```
MySQL 8.0 Command-Line Client > -- =====  
(PES1UG23C5924@root@localhost) [transit_gap] > -- =====  
(PES1UG23C5924@root@localhost) [transit_gap] > CREATE TABLE COMMUTER (  
->    Commuter_ID VARCHAR(12) NOT NULL,  
->    First_Name  VARCHAR(30) NOT NULL,  
->    Middle_Name VARCHAR(30) NULL,  
->    Last_Name   VARCHAR(30) NOT NULL,  
->    Gender      CHAR(1) NOT NULL,  
->    Date_of_Birth DATE NOT NULL,  
->    Email_Address VARCHAR(60) NOT NULL,  
->    Ward_ID     VARCHAR(10) NOT NULL,  
->    CONSTRAINT pk_commuter PRIMARY KEY (Commuter_ID),  
->    CONSTRAINT uq_commuter_email UNIQUE (Email_Address)  
-> );  
Query OK, 0 rows affected (0.02 sec)  
(PES1UG23C5924@root@localhost) [transit_gap] >  
(PES1UG23C5924@root@localhost) [transit_gap] > CREATE TABLE COMMUTER_PHONE (  
->    Commuter_ID VARCHAR(12) NOT NULL,  
->    Phone_Number VARCHAR(15) NOT NULL,  
->    CONSTRAINT pk_commuter_phone PRIMARY KEY (Commuter_ID, Phone_Number)  
-> );  
Query OK, 0 rows affected (0.01 sec)  
(PES1UG23C5924@root@localhost) [transit_gap] >  
(PES1UG23C5924@root@localhost) [transit_gap] > CREATE TABLE HOLDS (  
->    Commuter_ID VARCHAR(12) NOT NULL,  
->    Pass_Type_ID VARCHAR(18) NOT NULL,  
->    CONSTRAINT pk_holds PRIMARY KEY (Commuter_ID, Pass_Type_ID)  
-> );
```

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## 2.1e CREATE TABLE - one FARE\_RECEIPT per JOURNEY



```
CREATE TABLE JOURNEY (  
    Journey_ID    VARCHAR(12) NOT NULL,  
    Journey_Date  DATE        NOT NULL,  
    Commuter_ID   VARCHAR(12) NOT NULL,  
    CONSTRAINT pk_journey PRIMARY KEY (Journey_ID));
```

### UNIQUE on the foreign key is what allows only one

```
CREATE TABLE FARE_RECEIPT (  
    Receipt_ID    VARCHAR(12) NOT NULL,  
    Fare_Amount   DECIMAL(10,2) NOT NULL,  
    Journey_ID    VARCHAR(12) NOT NULL,  
    CONSTRAINT pk_fare_receipt PRIMARY KEY (Receipt_ID),  
    CONSTRAINT uq_receipt_journey UNIQUE (Journey_ID);
```

```
MySQL 8.0 Command Line Client  
(PES1UG23CS024@root@localhost) [transit_gap] > CREATE TABLE HOLDS (  
-> Commuter_ID VARCHAR(12) NOT NULL,  
-> Pass_Type_ID VARCHAR(10) NOT NULL,  
-> CONSTRAINT pk_holds PRIMARY KEY (Commuter_ID, Pass_Type_ID)  
-> );  
Query OK, 0 rows affected (0.01 sec)  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > -- 2.1e CREATE TABLE - the 1:1 pair JOURNEY and FARE_RECEIPT  
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====  
(PES1UG23CS024@root@localhost) [transit_gap] > -- UNIQUE on the foreign key is what makes GENERATES 1:1 and not 1:N  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > CREATE TABLE JOURNEY (  
-> Journey_ID    VARCHAR(12) NOT NULL,  
-> Journey_Date  DATE        NOT NULL,  
-> Start_Time    TIME        NOT NULL,  
-> End_Time      TIME        NULL,  
-> Transfer_Count INT        NOT NULL DEFAULT 0,  
-> Commuter_ID   VARCHAR(12) NOT NULL,  
-> CONSTRAINT pk_journey PRIMARY KEY (Journey_ID),  
-> CONSTRAINT chk_journey_transfers CHECK (Transfer_Count >= 0),  
-> CONSTRAINT chk_journey_times CHECK (End_Time IS NULL OR End_Time >= Start_Time)  
-> );  
Query OK, 0 rows affected (0.01 sec)  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > CREATE TABLE FARE_RECEIPT (  
-> Receipt_ID    VARCHAR(12) NOT NULL,  
-> Fare_Amount   DECIMAL(10,2) NOT NULL,  
-> Journey_ID    VARCHAR(12) NOT NULL,  
-> CONSTRAINT pk_fare_receipt PRIMARY KEY (Receipt_ID),  
-> CONSTRAINT uq_receipt_journey UNIQUE (Journey_ID),  
-> CONSTRAINT chk_receipt_amount CHECK (Fare_Amount >= 0)  
-> );  
Query OK, 0 rows affected (0.02 sec)
```

```
MySQL 8.0 Command Line Client  
(PES1UG23CS024@root@localhost) [transit_gap] > CREATE TABLE FARE_RECEIPT (  
-> Receipt_ID    VARCHAR(12) NOT NULL,  
-> Fare_Amount   DECIMAL(10,2) NOT NULL,  
-> Journey_ID    VARCHAR(12) NOT NULL,  
-> CONSTRAINT pk_fare_receipt PRIMARY KEY (Receipt_ID),  
-> CONSTRAINT uq_receipt_journey UNIQUE (Journey_ID),  
-> CONSTRAINT chk_receipt_amount CHECK (Fare_Amount >= 0)  
-> );  
Query OK, 0 rows affected (0.02 sec)  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > -- 2.1f Foreign keys - one referential action per statement  
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====  
(PES1UG23CS024@root@localhost) [transit_gap] > ALTER TABLE ROUTE  
-> ADD CONSTRAINT fk_route_agency FOREIGN KEY (Agency_ID)  
-> REFERENCES AGENCY (Agency_ID) ON UPDATE CASCADE;  
Query OK, 0 rows affected (0.05 sec)  
Records: 0 Duplicates: 0 Warnings: 0  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > -- Recursive FEEDS. Never CASCADE here: withdrawing one trunk route  
(PES1UG23CS024@root@localhost) [transit_gap] > -- would delete every feeder route pointing at it.  
(PES1UG23CS024@root@localhost) [transit_gap] > ALTER TABLE ROUTE  
-> ADD CONSTRAINT fk_route_trunk FOREIGN KEY (Trunk_Route_ID)  
-> REFERENCES ROUTE (Route_ID) ON DELETE SET NULL;  
Query OK, 0 rows affected (0.04 sec)
```

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## 2.1f Foreign keys - one referential action at a time

```
ALTER TABLE ROUTE ADD CONSTRAINT fk_route_agency
    FOREIGN KEY (Agency_ID) REFERENCES AGENCY (Agency_ID)
    ON UPDATE CASCADE;
```

```
ALTER TABLE ROUTE ADD CONSTRAINT fk_route_trunk
    FOREIGN KEY (Trunk_Route_ID) REFERENCES ROUTE (Route_ID)
    ON DELETE SET NULL;      -- never CASCADE here
```

```
ALTER TABLE STOP ADD CONSTRAINT fk_stop_agency
    FOREIGN KEY (Agency_ID) REFERENCES AGENCY (Agency_ID)
    ON DELETE CASCADE;      -- STOP cannot exist alone
```

```
ALTER TABLE STOP ADD CONSTRAINT fk_stop_ward
    FOREIGN KEY (Ward_ID) REFERENCES WARD (Ward_ID)
    ON UPDATE CASCADE;
```

```
MySQL 8.0 Command Line CL >
(PES1UG23CS024@rootlocalhost) [transit_gap] > -- Recursive FEEDS. Never CASCADE here; withdrawing one trunk route
(PES1UG23CS024@rootlocalhost) [transit_gap] > -- would delete every feeder route pointing at it.
(PES1UG23CS024@rootlocalhost) [transit_gap] > ALTER TABLE ROUTE
-> ADD CONSTRAINT fk_route_trunk FOREIGN KEY (Trunk_Route_ID)
-> REFERENCES ROUTE (Route_ID) ON DELETE SET NULL;
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@rootlocalhost) [transit_gap] > -- Identifying relationship: a stop cannot outlive its agency.
(PES1UG23CS024@rootlocalhost) [transit_gap] > ALTER TABLE STOP
-> ADD CONSTRAINT fk_stop_agency FOREIGN KEY (Agency_ID)
-> REFERENCES AGENCY (Agency_ID) ON DELETE CASCADE;
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@rootlocalhost) [transit_gap] >
(PES1UG23CS024@rootlocalhost) [transit_gap] > ALTER TABLE STOP
-> ADD CONSTRAINT fk_stop_ward FOREIGN KEY (Ward_ID)
-> REFERENCES WARD (Ward_ID) ON UPDATE CASCADE;
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@rootlocalhost) [transit_gap] >
(PES1UG23CS024@rootlocalhost) [transit_gap] > ALTER TABLE PLATFORM
-> ADD CONSTRAINT fk_platform_stop FOREIGN KEY (Agency_ID, Stop_Code)
-> REFERENCES STOP (Agency_ID, Stop_Code) ON DELETE CASCADE;
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
MySQL 8.0 Command Line CL >
(PES1UG23CS024@rootlocalhost) [transit_gap] >
(PES1UG23CS024@rootlocalhost) [transit_gap] > ALTER TABLE SERVES
-> ADD CONSTRAINT fk_serves_route FOREIGN KEY (Route_ID)
-> REFERENCES ROUTE (Route_ID) ON DELETE CASCADE;
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@rootlocalhost) [transit_gap] >
(PES1UG23CS024@rootlocalhost) [transit_gap] > ALTER TABLE SERVES
-> ADD CONSTRAINT fk_serves_stop FOREIGN KEY (Agency_ID, Stop_Code)
-> REFERENCES STOP (Agency_ID, Stop_Code) ON DELETE CASCADE;
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@rootlocalhost) [transit_gap] >
(PES1UG23CS024@rootlocalhost) [transit_gap] > ALTER TABLE PASS_TYPE
-> ADD CONSTRAINT fk_pass_agency FOREIGN KEY (Agency_ID)
-> REFERENCES AGENCY (Agency_ID) ON DELETE CASCADE;
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@rootlocalhost) [transit_gap] >
(PES1UG23CS024@rootlocalhost) [transit_gap] > ALTER TABLE COMMUTER
-> ADD CONSTRAINT fk_commuter_ward FOREIGN KEY (Ward_ID)
-> REFERENCES WARD (Ward_ID) ON UPDATE CASCADE;
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@rootlocalhost) [transit_gap] >
```

```
MySQL 8.0 Command Line CL >
(PES1UG23CS024@rootlocalhost) [transit_gap] > ALTER TABLE COMMUTER_PHONE
-> ADD CONSTRAINT fk_phone_commuter FOREIGN KEY (Commuter_ID)
-> REFERENCES COMMUTER (Commuter_ID) ON DELETE CASCADE;
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@rootlocalhost) [transit_gap] >
(PES1UG23CS024@rootlocalhost) [transit_gap] > ALTER TABLE HOLDS
-> ADD CONSTRAINT fk_holds_commuter FOREIGN KEY (Commuter_ID)
-> REFERENCES COMMUTER (Commuter_ID) ON DELETE CASCADE;
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@rootlocalhost) [transit_gap] >
(PES1UG23CS024@rootlocalhost) [transit_gap] > ALTER TABLE HOLDS
-> ADD CONSTRAINT fk_holds_pass FOREIGN KEY (Pass_Type_ID)
-> REFERENCES PASS_TYPE (Pass_Type_ID) ON DELETE CASCADE;
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@rootlocalhost) [transit_gap] >
(PES1UG23CS024@rootlocalhost) [transit_gap] > ALTER TABLE JOURNEY
-> ADD CONSTRAINT fk_journey_commuter FOREIGN KEY (Commuter_ID)
-> REFERENCES COMMUTER (Commuter_ID) ON DELETE CASCADE;
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

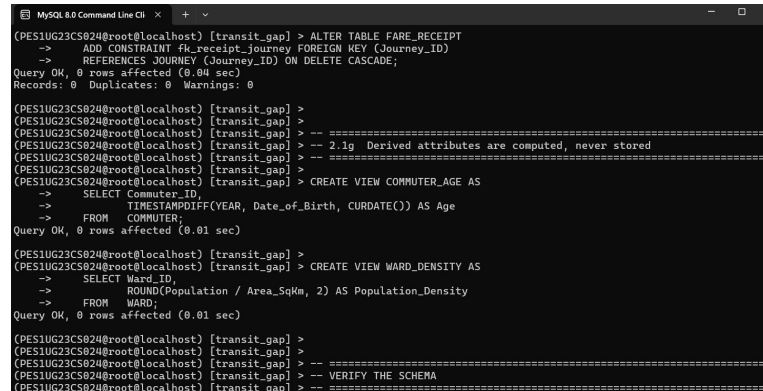
(PES1UG23CS024@rootlocalhost) [transit_gap] >
(PES1UG23CS024@rootlocalhost) [transit_gap] > ALTER TABLE FARE_RECEIPT
-> ADD CONSTRAINT fk_receipt_journey FOREIGN KEY (Journey_ID)
```

### Age and Population\_Density are computed from other columns

```
CREATE VIEW COMMUTER_AGE AS
    SELECT Commuter_ID,
           TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
    FROM   COMMUTER;

CREATE VIEW WARD_DENSITY AS
    SELECT Ward_ID, ROUND(Population / Area_SqKm, 2) AS Population_Density
    FROM   WARD;

SELECT * FROM WARD_DENSITY;
```



```
MySQL 8.0 Command Line Client
(PESIUG23CS024@root@localhost) [transit_gap] > ALTER TABLE FARE_RECEIPT
-- ADD CONSTRAINT fk_receipt_journey FOREIGN KEY (Journey_ID)
-- REFERENCES JOURNEY (Journey_ID) ON DELETE CASCADE;
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PESIUG23CS024@root@localhost) [transit_gap] >
(PESIUG23CS024@root@localhost) [transit_gap] >
(PESIUG23CS024@root@localhost) [transit_gap] > -- 2.1g Derived attributes are computed, never stored
(PESIUG23CS024@root@localhost) [transit_gap] > --
(PESIUG23CS024@root@localhost) [transit_gap] >
(PESIUG23CS024@root@localhost) [transit_gap] > CREATE VIEW COMMUTER_AGE AS
-- SELECT Commuter_ID,
--     TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
-- FROM   COMMUTER;
Query OK, 0 rows affected (0.01 sec)

(PESIUG23CS024@root@localhost) [transit_gap] >
(PESIUG23CS024@root@localhost) [transit_gap] > CREATE VIEW WARD_DENSITY AS
-- SELECT Ward_ID,
--     ROUND(Population / Area_SqKm, 2) AS Population_Density
-- FROM   WARD;
Query OK, 0 rows affected (0.01 sec)

(PESIUG23CS024@root@localhost) [transit_gap] >
(PESIUG23CS024@root@localhost) [transit_gap] >
(PESIUG23CS024@root@localhost) [transit_gap] > --
(PESIUG23CS024@root@localhost) [transit_gap] > -- VERIFY THE SCHEMA
(PESIUG23CS024@root@localhost) [transit_gap] > --
```



# Database Management Systems

## Verify the schema



```
SHOW TABLES;
```

```
DESC AGENCY;
```

```
DESC STOP;
```

```
SHOW CREATE TABLE STOP;
```

```
-- confirm: PRIMARY KEY (Agency_ID, Stop_Code) and Ward_ID NOT NULL
```

```
MySQL 8.0 Command Line Client >
(PES1UG23CS824@rootlocalhost) [transit_gap] > DESC AGENCY;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| Agency_ID | varchar(10) | NO | PRI | NULL | |
| Agency_Name | varchar(60) | NO | UNI | NULL | |
| Route | varchar(10) | NO | | NULL | |
| Headquarters_Location | varchar(100) | NO | NULL | | |
| Fleet_Size | int | NO | 0 | | |
+-----+
5 rows in set (0.00 sec)

(PES1UG23CS824@rootlocalhost) [transit_gap] > DESC WARD;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| Ward_ID | varchar(10) | NO | PRI | NULL | |
| Ward_Name | varchar(60) | NO | NULL | UNASSIGNED | |
| Zone | varchar(40) | NO | | NULL | |
| Population | int | NO | 0 | | |
| Area_SqM | decimal(8,2) | NO | NULL | | |
+-----+
5 rows in set (0.00 sec)

(PES1UG23CS824@rootlocalhost) [transit_gap] > DESC ROUTE;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| Route_ID | varchar(10) | NO | PRI | NULL | |
| Route_Number | varchar(12) | NO | | NULL | |
+-----+
2 rows in set (0.00 sec)
```

```
MySQL 8.0 Command Line Client >
(PES1UG23CS824@rootlocalhost) [transit_gap] > DESC ROUTE;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| Route_ID | varchar(10) | NO | PRI | NULL | |
| Route_Number | varchar(12) | NO | | NULL | |
| Route_Name | varchar(60) | NO | | NULL | |
| Route_Type | varchar(20) | NO | | NULL | |
| Total_Distance_KM | decimal(6,2) | NO | | NULL | |
| Agency_ID | varchar(10) | NO | | NULL | |
| Trunk_Route_ID | varchar(10) | YES | | NULL | |
+-----+
7 rows in set (0.00 sec)

(PES1UG23CS824@rootlocalhost) [transit_gap] > DESC STOP;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| Agency_ID | varchar(10) | NO | PRI | NULL | |
| Stop_Code | varchar(12) | NO | PRI | NULL | |
| Stop_Name | varchar(60) | NO | | NULL | |
| Latitude | decimal(9,6) | NO | | NULL | |
| Longitude | decimal(9,6) | NO | | NULL | |
| Ward_ID | varchar(10) | NO | | NULL | |
+-----+
6 rows in set (0.00 sec)

(PES1UG23CS824@rootlocalhost) [transit_gap] > DESC PLATFORM;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| Agency_ID | varchar(10) | NO | PRI | NULL | |
| Platform_Number | varchar(5) | NO | PRI | NULL | |
| Wheelchair_Access | char(1) | NO | | N | |
+-----+
4 rows in set (0.00 sec)
```

```
MySQL 8.0 Command Line Client >
(PES1UG23CS824@rootlocalhost) [transit_gap] > DESC PLATFORM;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| Agency_ID | varchar(10) | NO | PRI | NULL | |
| Stop_Code | varchar(12) | NO | PRI | NULL | |
| Platform_Number | varchar(5) | NO | PRI | NULL | |
| Wheelchair_Access | char(1) | NO | | N | |
+-----+
4 rows in set (0.00 sec)

(PES1UG23CS824@rootlocalhost) [transit_gap] > DESC SERVES;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| Route_ID | varchar(10) | NO | PRI | NULL | |
| Agency_ID | varchar(10) | NO | PRI | NULL | |
| Stop_Code | varchar(12) | NO | PRI | NULL | |
| Stop_Sequence | int | NO | | NULL | |
| Scheduled_Offset_Minutes | int | NO | | 0 | |
| Distance_From_Origin_KM | decimal(6,2) | NO | | 0.00 | |
+-----+
6 rows in set (0.00 sec)

(PES1UG23CS824@rootlocalhost) [transit_gap] > DESC PASS_TYPE;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| Pass_Type_ID | varchar(10) | NO | PRI | NULL | |
+-----+
1 row in set (0.00 sec)
```

```
MySQL 8.0 Command Line Client >
(PES1UG23CS824@rootlocalhost) [transit_gap] > DESC PASS_TYPE;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| Pass_Type_ID | varchar(10) | NO | PRI | NULL | |
| Pass_Name | varchar(20) | YES | | NULL | |
| Validity_Days | int | NO | | NULL | |
| Price | decimal(10,2) | NO | | NULL | |
| Agency_ID | varchar(10) | NO | | NULL | |
+-----+
5 rows in set (0.00 sec)

(PES1UG23CS824@rootlocalhost) [transit_gap] > DESC COMPUTER;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| Computer_ID | varchar(12) | NO | PRI | NULL | |
| First_Name | varchar(20) | NO | | NULL | |
| Last_Name | varchar(20) | NO | | NULL | |
| Gender | char(1) | NO | | NULL | |
| Date_of_Birth | date | NO | | NULL | |
| Email_Address | varchar(60) | NO | | NULL | |
| Marry_ID | varchar(10) | NO | | NULL | |
+-----+
7 rows in set (0.00 sec)

(PES1UG23CS824@rootlocalhost) [transit_gap] > DESC COMPUTER_PHONE;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| Computer_ID | varchar(12) | NO | PRI | NULL | |
| Phone_Number | varchar(15) | NO | | NULL | |
+-----+
2 rows in set (0.00 sec)
```

```
MySQL 8.0 Command Line Client >
(PES1UG23CS824@rootlocalhost) [transit_gap] > DESC FARE_RECEIPT;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| Receipt_ID | varchar(12) | NO | PRI | NULL | |
| Receipt_Date | date | NO | | NULL | |
| Fare_Amount | decimal(10,2) | NO | | NULL | |
| Journey_ID | varchar(12) | NO | | NULL | |
+-----+
4 rows in set (0.00 sec)

(PES1UG23CS824@rootlocalhost) [transit_gap] >
-- Confirm: PRIMARY KEY (Agency_ID, Stop_Code) and Ward_ID
(PES1UG23CS824@rootlocalhost) [transit_gap] > SHOW CREATE TABLE STOP;
+-----+
| Table | Create Table
+-----+
| Table | Create Table
+-----+
1 row in set (0.01 sec)
```

```
MySQL 8.0 Command Line Client >
(PES1UG23CS824@rootlocalhost) [transit_gap] > SHOW CREATE TABLE STOP;
+-----+
| Table | Create Table
+-----+
| STOP | CREATE TABLE `stop` (
  `Agency_ID` varchar(10) NOT NULL,
  `Stop_Code` varchar(12) NOT NULL,
  `Stop_Name` varchar(60) NOT NULL,
  `Latitude` decimal(9,6) NOT NULL,
  `Longitude` decimal(9,6) NOT NULL,
  `Ward_ID` varchar(10) NOT NULL,
  PRIMARY KEY (`Agency_ID`,`Stop_Code`),
  UNIQUE INDEX (`Ward_ID`),
  CONSTRAINT `fk_stop_agency` FOREIGN KEY (`Agency_ID`) REFERENCES `agency` (`Agency_ID`) ON DELETE CASCADE,
  CONSTRAINT `chk_stop_lat` CHECK ((Latitude between -90 and 90)),
  CONSTRAINT `chk_stop_lon` CHECK ((Longitude between -180 and 180))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci
+-----+
1 row in set (0.01 sec)
```

```
MySQL 8.0 Command Line Client >
(PES1UG23CS824@rootlocalhost) [transit_gap] > SHOW CREATE TABLE ROUTE;
+-----+
| Table | Create Table
+-----+
| ROUTE | CREATE TABLE `route` (
  `Route_ID` varchar(10) NOT NULL,
  `Route_Number` varchar(12) NOT NULL,
  `Route_Name` varchar(60) NOT NULL,
  PRIMARY KEY (`Route_ID`,`Route_Number`),
  UNIQUE INDEX (`Route_Name`),
  CONSTRAINT `fk_route_agency` FOREIGN KEY (`Agency_ID`) REFERENCES `agency` (`Agency_ID`) ON DELETE CASCADE,
  CONSTRAINT `chk_route_lat` CHECK ((Latitude between -90 and 90)),
  CONSTRAINT `chk_route_lon` CHECK ((Longitude between -180 and 180))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci
+-----+
1 row in set (0.01 sec)
```

# Database Management Systems

## Verify the schema - SHOW CREATE TABLE reads it back

```
MySQL 8.0 Command Line CL>
+-----+
| ROUTE | CREATE TABLE `route` (
  `Route_ID` varchar(10) NOT NULL,
  `Route_Number` varchar(15) NOT NULL,
  `Route_Name` varchar(80) NOT NULL,
  `Route_Type` varchar(20) NOT NULL DEFAULT 'ORDINARY',
  `Total_Distance_KM` decimal(6,2) NOT NULL,
  `Agency_ID` varchar(10) NOT NULL,
  `Trunk_Route_ID` varchar(10) DEFAULT NULL,
  PRIMARY KEY (`Route_ID`),
  KEY `fk_route_agency` (`Agency_ID`),
  KEY `fk_route_trunk` (`Trunk_Route_ID`),
  CONSTRAINT `fk_route_agency` FOREIGN KEY (`Agency_ID`) REFERENCES `agency` (`Agency_ID`) ON UPDATE CASCADE,
  CONSTRAINT `fk_route_trunk` FOREIGN KEY (`Trunk_Route_ID`) REFERENCES `route` (`Route_ID`) ON DELETE SET NULL,
  CONSTRAINT `chk_route_distance` CHECK ((`Total_Distance_KM` > 0))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+

1 row in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] > SHOW CREATE TABLE FARE_RECEIPT;
+-----+
| Table | Create Table
+-----+
|
+-----+
|
+-----+
| FARE_RECEIPT | CREATE TABLE `fare_receipt` (
  `Receipt_ID` varchar(12) NOT NULL,
  `Receipt_Date` date NOT NULL,
  `Fare_Amount` decimal(10,2) NOT NULL,
  `Journey_ID` varchar(12) NOT NULL,
  PRIMARY KEY (`Receipt_ID`),
  UNIQUE KEY `uq_receipt_journey` (`Journey_ID`),
  CONSTRAINT `fk_receipt_journey` FOREIGN KEY (`Journey_ID`) REFERENCES `journey` (`Journey_ID`) ON DELETE CASCADE,
  CONSTRAINT `chk_receipt_amount` CHECK ((`Fare_Amount` >= 0))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+
```

```
MySQL 8.0 Command Line CL>
+-----+
| Table | Create Table
+-----+
|
+-----+
|
+-----+
| FARE_RECEIPT | CREATE TABLE `fare_receipt` (
  `Receipt_ID` varchar(12) NOT NULL,
  `Receipt_Date` date NOT NULL,
  `Fare_Amount` decimal(10,2) NOT NULL,
  `Journey_ID` varchar(12) NOT NULL,
  PRIMARY KEY (`Receipt_ID`),
  UNIQUE KEY `uq_receipt_journey` (`Journey_ID`),
  CONSTRAINT `fk_receipt_journey` FOREIGN KEY (`Journey_ID`) REFERENCES `journey` (`Journey_ID`) ON DELETE CASCADE,
  CONSTRAINT `chk_receipt_amount` CHECK ((`Fare_Amount` >= 0))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+
```

The two that carry every referential action declared in step 2.1f

ROUTE shows `fk_route_agency` ON UPDATE CASCADE and `fk_route_trunk` ON DELETE SET NULL.

FARE\_RECEIPT shows `uq_receipt_journey` - the UNIQUE that allows one receipt per journey.

## 2.2 ALTER TABLE - add a column as the first column

```
ALTER TABLE PASS_TYPE
    ADD COLUMN Pass_Category VARCHAR(15) NOT NULL DEFAULT 'GENERAL' FIRST;

DESC PASS_TYPE;
```

```
MySQL 8.0 Command Line Client
1 row in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- 2.2 ALTER TABLE - add a new attribute as the FIRST column
(PES1UG23CS024@root@localhost) [transit_gap] > --
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > ALTER TABLE PASS_TYPE
    > ADD COLUMN Pass_Category VARCHAR(15) NOT NULL DEFAULT 'GENERAL' FIRST;
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > DESC PASS_TYPE;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Pass_Category | varchar(15) | NO   |     | GENERAL |       |
| Pass_Type_ID | varchar(10) | NO   | PRI | NULL    |       |
| Pass_Name    | varchar(50) | NO   |     | NULL    |       |
| Validity_Days | int        | NO   |     | 1       |       |
| Price        | decimal(10,2) | NO   |     | NULL    |       |
| Agency_ID    | varchar(10) | NO   | MUL | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
```

# Database Management Systems

## 2.3 ALTER TABLE - add a missed constraint



```
ALTER TABLE PASS_TYPE
    ADD CONSTRAINT uq_pass_name UNIQUE (Pass_Name);
```

```
ALTER TABLE PASS_TYPE
    ADD CONSTRAINT chk_pass_category
    CHECK (CHAR_LENGTH(Pass_Category) > 0);
```

```
SHOW CREATE TABLE PASS_TYPE;
```

```
MySQL 8.0 Command Line CLI
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- 2.3 ALTER TABLE - add constraints missed at create time
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > ALTER TABLE PASS_TYPE
->
    ADD CONSTRAINT uq_pass_name UNIQUE (Pass_Name);
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > ALTER TABLE PASS_TYPE
->
    ADD CONSTRAINT chk_pass_category
    CHECK (CHAR_LENGTH(Pass_Category) > 0);
Query OK, 0 rows affected (0.00 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SHOW CREATE TABLE PASS_TYPE;
+-----+
| Table | Create Table
+-----+
+-----+
1 row in set (0.00 sec)
```

```
MySQL 8.0 Command Line CLI
+-----+
| PASS_TYPE | CREATE TABLE 'pass_type' (
  'Pass_Category' varchar(15) NOT NULL DEFAULT 'GENERAL',
  'Pass_Type_ID' varchar(10) NOT NULL,
  'Pass_Name' varchar(50) NOT NULL,
  'Validity_Days' int NOT NULL DEFAULT '1',
  'Price' decimal(10,2) NOT NULL,
  'Agency_ID' varchar(10) NOT NULL,
  PRIMARY KEY ('Pass_Type_ID'),
  UNIQUE KEY 'uq_pass_name' ('Pass_Name'),
  KEY 'fk_pass_agency' ('Agency_ID'),
  CONSTRAINT 'fk_pass_agency' FOREIGN KEY ('Agency_ID') REFERENCES 'agency' ('Agency_ID') ON DELETE CASCADE,
  CONSTRAINT 'chk_pass_category' CHECK ((char_length('Pass_Category') > 0)),
  CONSTRAINT 'chk_pass_price' CHECK ((('Price' >= 0)),
  CONSTRAINT 'chk_pass_validity' CHECK ((('Validity_Days' > 0)))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+
1 row in set (0.00 sec)
```

# Database Management Systems

## 2.4 / 2.5 ALTER TABLE - rename a column, change a datatype



### 2.4 Rename a column

```
ALTER TABLE ROUTE RENAME COLUMN Route_Type TO Category;
```

```
DESC ROUTE; -- MySQL 5.7 and earlier use CHANGE COLUMN
```

### 2.5 Change a datatype

```
ALTER TABLE COMMUTER MODIFY COLUMN Gender ENUM('M','F','O') NOT NULL;
```

```
DESC COMMUTER;
```

```
MySQL 8.0 Command Line CLI [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- 2.4 ALTER TABLE - rename a column
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > ALTER TABLE ROUTE RENAME COLUMN Route_Type TO Category;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > DESC ROUTE;
+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+
| Route_ID | varchar(10) | NO | PRI | NULL | |
| Route_Number | varchar(15) | NO | | NULL | |
| Route_Name | varchar(80) | NO | | NULL | |
| Category | varchar(20) | NO | | ORDINARY | |
| Total_Distance_KM | decimal(6,2) | NO | | NULL | |
| Agency_ID | varchar(10) | NO | MUL | NULL | |
| Trunk_Route_ID | varchar(10) | YES | MUL | NULL | |
+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- MySQL 5.7 and earlier use CHANGE COLUMN instead:
(PES1UG23CS024@root@localhost) [transit_gap] > -- ALTER TABLE ROUTE CHANGE COLUMN Route_Type Category
(PES1UG23CS024@root@localhost) [transit_gap] > -- VARCHAR(20) NOT NULL DEFAULT 'ORDINARY';
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
```

```
MySQL 8.0 Command Line CLI [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- 2.5 ALTER TABLE - change the datatype from CHAR to ENUM
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > ALTER TABLE COMMUTER
-> MODIFY COLUMN Gender ENUM('M','F','O') NOT NULL;
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > DESC COMMUTER;
+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+
| Commuter_ID | varchar(12) | NO | PRI | NULL | |
| First_Name | varchar(30) | NO | | NULL | |
| Middle_Name | varchar(30) | YES | | NULL | |
| Last_Name | varchar(30) | NO | | NULL | |
| Gender | enum('M','F','O') | NO | | NULL | |
| Date_of_Birth | date | NO | | NULL | |
| Email_Address | varchar(60) | NO | UNI | NULL | |
| Ward_ID | varchar(10) | NO | MUL | NULL | |
+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)
```

## 2.6 ALTER TABLE - drop the demonstration column

### Drop the named constraint first, or the column will not drop

```
ALTER TABLE PASS_TYPE DROP CONSTRAINT chk_pass_category;
```

```
ALTER TABLE PASS_TYPE RENAME COLUMN Pass_Category TO Pass_Tag;
```

```
ALTER TABLE PASS_TYPE DROP COLUMN Pass_Tag;
```

```
DESC PASS_TYPE;
```

```
MySQL 8.0 Command Line C> [transit_gap] > -- =====
[transit_gap] > -- 2.6 ALTER TABLE - drop the column added in step 2.2
[transit_gap] > -- Uncomment the next line to demonstrate the failure first.
[transit_gap] > -- It fails because chk_pass_category still references the column.
[transit_gap] > -- ALTER TABLE PASS_TYPE DROP COLUMN Pass_Category;
[transit_gap] > ALTER TABLE PASS_TYPE DROP CONSTRAINT chk_pass_category;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

[transit_gap] >
[transit_gap] > ALTER TABLE PASS_TYPE RENAME COLUMN Pass_Category TO Pass_Tag;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

[transit_gap] >
[transit_gap] > ALTER TABLE PASS_TYPE DROP COLUMN Pass_Tag;
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0

[transit_gap] >
[transit_gap] > DESC PASS_TYPE;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| Pass_Type_ID | varchar(10) | NO | PRI | NULL | |
| Pass_Name | varchar(50) | NO | UNI | NULL | |
| Validity_Days | int | NO | 1 | NULL | |
| Price | decimal(10,2) | NO | | NULL | |
+-----+
```

```
MySQL 8.0 Command Line C> [transit_gap] > DESC PASS_TYPE;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| Pass_Type_ID | varchar(10) | NO | PRI | NULL | |
| Pass_Name | varchar(50) | NO | UNI | NULL | |
| Validity_Days | int | NO | 1 | NULL | |
| Price | decimal(10,2) | NO | | NULL | |
| Agency_ID | varchar(10) | NO | MUL | NULL | |
+-----+
5 rows in set (0.00 sec)

[transit_gap] >
[transit_gap] > -- Before MySQL 8.0.19 use these two forms instead:
[transit_gap] > -- ALTER TABLE PASS_TYPE DROP CHECK chk_pass_category;
[transit_gap] > -- ALTER TABLE PASS_TYPE DROP INDEX uq_pass_name;
[transit_gap] >
[transit_gap] >
[transit_gap] > -- =====
[transit_gap] > -- 2.7 ALTER TABLE - change a column's size
[transit_gap] > -- =====
[transit_gap] >
[transit_gap] > ALTER TABLE COMMUTER
-> MODIFY COLUMN Email_Address VARCHAR(120) NOT NULL;
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0

[transit_gap] >
[transit_gap] > ALTER TABLE STOP
-> MODIFY COLUMN Stop_Name VARCHAR(120) NOT NULL;
```

# Database Management Systems

## 2.7 ALTER TABLE - change a column's size



```
ALTER TABLE COMMUTER
```

```
    MODIFY COLUMN Email_Address VARCHAR(120) NOT NULL;
```

```
ALTER TABLE STOP
```

```
    MODIFY COLUMN Stop_Name VARCHAR(120) NOT NULL;
```

```
DESC COMMUTER;
```

```
MySQL 8.0 Command Line CL> + -
(PES1UG23CS824@root@localhost) [transit_gap] > ALTER TABLE STOP
>    MODIFY COLUMN Stop_Name VARCHAR(120) NOT NULL;
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES1UG23CS824@root@localhost) [transit_gap] >
(PES1UG23CS824@root@localhost) [transit_gap] > DESC COMMUTER;
+----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+----+-----+-----+-----+-----+-----+
| Commuter_ID | varchar(12) | NO | PRI | NULL | |
| First_Name | varchar(30) | NO | | NULL | |
| Middle_Name | varchar(30) | YES | | NULL | |
| Last_Name | varchar(30) | NO | | NULL | |
| Gender | enum('M','F','O') | NO | | NULL | |
| Date_of_Birth | date | NO | | NULL | |
| Email_Address | varchar(120) | NO | UNI | NULL | |
| Ward_ID | varchar(10) | NO | MUL | NULL | |
+----+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)

(PES1UG23CS824@root@localhost) [transit_gap] > DESC STOP;
+----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+----+-----+-----+-----+-----+-----+
| Agency_ID | varchar(10) | NO | PRI | NULL | |
| Stop_Code | varchar(12) | NO | PRI | NULL | |
| Stop_Name | varchar(120) | NO | | NULL | |
| Latitude | decimal(9,6) | NO | | NULL | |
| Longitude | decimal(9,6) | NO | | NULL | |
+----+-----+-----+-----+-----+-----+
```

```
ALTER TABLE COMMUTER RENAME TO COMMUTERS;
```

```
SHOW TABLES;
```

```
SHOW CREATE TABLE JOURNEY;
```

```
-- InnoDB has rewritten the reference to COMMUTERS automatically
```

```
ALTER TABLE COMMUTERS RENAME TO COMMUTER;
```

```
> MySQL Command Line C# >
(PESUIG2JC9S0Z@root)[localhost] [transit_gap] > DESC STOP;
+-----+
| Field | Type                | Null | Key | Default | Extra |
+-----+
| Agency_ID | varchar(12)         | NO   | PRI | NULL    |       |
| Stop_Code | varchar(12)         | NO   | PRI | NULL    |       |
| Stop_Name | varchar(128)        | NO   |     | NULL    |       |
| Latitude  | decimal(9,6)        | NO   |     | NULL    |       |
| Longitude | decimal(9,6)        | NO   |     | NULL    |       |
| Ward_ID   | varchar(18)         | NO   | MUL | NULL    |       |
+-----+
6 rows in set (0.00 sec)

(PESUIG2JC9S0Z@root)[localhost] [transit_gap] >
(PESUIG2JC9S0Z@root)[localhost] [transit_gap] >
(PESUIG2JC9S0Z@root)[localhost] [transit_gap] >
(PESUIG2JC9S0Z@root)[localhost] [transit_gap] -- 2.8 RENAME TABLE - what happens to the foreign keys?
(PESUIG2JC9S0Z@root)[localhost] [transit_gap] > ALTER TABLE COMPUTERS RENAME TO COMPUTERS;
(PESUIG2JC9S0Z@root)[localhost] [transit_gap] >
Query OK, 0 rows affected (0.82 sec)

(PESUIG2JC9S0Z@root)[localhost] [transit_gap] >
(PESUIG2JC9S0Z@root)[localhost] [transit_gap] > SHOW TABLES;
+-----+
| Tables_in_transit_gap |
+-----+
| agency                 |
| transit_gap            |
```

```

JOURNEY | CREATE TABLE 'Journey' (
  Journey_ID varchar(12) NOT NULL,
  Journey_Date date NOT NULL,
  Start_Time time NOT NULL,
  End_Time time DEFAULT NULL,
  Transfer_Count INT NOT NULL DEFAULT '0',
  Computer_ID varchar(12) NOT NULL,
  PRIMARY KEY (Journey_ID)
KEY %$-journey_computer' ('Computer_ID'),
CONSTRAINT %$-journey_computer FOREIGN KEY ('Computer_ID') REFERENCES 'computers' ('Computer_ID') ON DELETE CASCADE,
CONSTRAINT %$-journey_time CHECK ((('End_Time' IS NULL) <> ('Start_Time' <> ''))),
CONSTRAINT %$-journey_transfer CHECK (('Transfer_Count' >= 0))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci

```

```

PS100132C824@root(localhost) [transit_gap] > SHOW TABLES;

+-----+
| Tables_in_transit_gap |
+-----+
| agency                |
| computer_age          |
| computer_phone        |
| computers              |
| fare_receipt          |
| helen                 |
| journey               |
| pass_type             |
| platform              |
| route                |
| serves                |
| stop                  |
| ward                  |
| ward_density          |
+-----+

14 rows in set (0.00 sec)

PS100132C824@root(localhost) [transit_gap] >
PS100132C824@root(localhost) [transit_gap] > --JmooD has rewritten the reference to COMPUTERS automatically
PS100132C824@root(localhost) [transit_gap] > SHOW CREATE TABLE JOURNEY;

+-----+
| Table |
+-----+
|       |
+-----+

```

```
MySQL 5.6 Command Line [0] >
1 row in set (0.00 sec)

(PES1U023C526@root[localhost] [transit_gpp] > ALTER TABLE COMPUTERS RENAME TO COMPUTER;
Query OK, 0 rows affected (0.02 sec)

(PES1U023C526@root[localhost] [transit_gpp] >
(PES1U023C526@root[localhost] [transit_gpp] > SHOW TABLES;
+-----+
| Tables_in_transit_gpp |
+-----+
| agency                |
| computer              |
| computer_age          |
| computer_phone        |
| fare_receipt          |
| helix                 |
| journey              |
| pass_type             |
| platform              |
| route                |
| services              |
| stop                 |
| ward                 |
| ward_density          |
+-----+
```



# Database Management Systems

## Referential actions in effect



```
INSERT INTO AGENCY VALUES ('BMTC','BMTC','BUS','Shanthinagar',6000);
INSERT INTO WARD VALUES ('W150','Jayanagar',3.400);
INSERT INTO STOP VALUES ('BMTC','S001','Jayanagar 4th Block','W150');

DELETE FROM AGENCY WHERE Agency_ID = 'BMTC';
SELECT * FROM STOP; -- empty: ON DELETE CASCADE removed the child
```

```
MySQL 8.0 Command Line CLI
(PES1UG23CS024@root@localhost) [transit_gap] > -- REFERENCE ACTIONS IN EFFECT (ON DELETE CASCADE)
(PES1UG23CS024@root@localhost) [transit_gap] > -- REFERENCE ACTIONS IN EFFECT (ON DELETE CASCADE)
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO WARD (Ward_ID, Ward_Name, Zone, Population, Area_SqKm)
-> VALUES ('W150', 'Jayanagar', 'South', 42000, 3.400);
Query OK, 1 row affected (0.01 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO AGENCY (Agency_ID, Agency_Name, Mode, Headquarters_Location,
Fleet_Size)
-> VALUES ('BMTC', 'Bengaluru Metropolitan Transport Corporation',
-> 'BUS', 'Shanthinagar', 6000);
Query OK, 1 row affected (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO STOP (Agency_ID, Stop_Code, Stop_Name, Latitude, Longitude, W
ard_ID)
-> VALUES ('BMTC', 'S001', 'Jayanagar 4th Block', 12.925100, 77.583400, 'W150');
Query OK, 1 row affected (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM STOP;
+-----+-----+-----+-----+-----+
| Agency_ID | Stop_Code | Stop_Name | Latitude | Longitude | Ward_ID |
+-----+-----+-----+-----+-----+
| BMTC      | S001      | Jayanagar 4th Block | 12.925100 | 77.583400 | W150    |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
MySQL 8.0 Command Line CLI
(PES1UG23CS024@root@localhost) [transit_gap] > -- Deleting the parent cascades to the child, because
(PES1UG23CS024@root@localhost) [transit_gap] > -- fk_stop_agency was declared ON DELETE CASCADE.
(PES1UG23CS024@root@localhost) [transit_gap] > DELETE FROM AGENCY WHERE Agency_ID = 'BMTC';
Query OK, 1 row affected (0.01 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM STOP; -- expect: Empty set
Empty set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM WARD; -- the ward survives: no cascade on that side
+-----+-----+-----+-----+-----+
| Ward_ID | Ward_Name | Zone | Population | Area_SqKm |
+-----+-----+-----+-----+-----+
| W150    | Jayanagar | South | 42000      | 3.400      |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM WARD_DENSITY;
+-----+-----+
| Ward_ID | Population_Density |
+-----+-----+
| W150    | 12352.94           |
+-----+-----+
1 row in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- 2.9 DROP - constraints first, then the tables
```

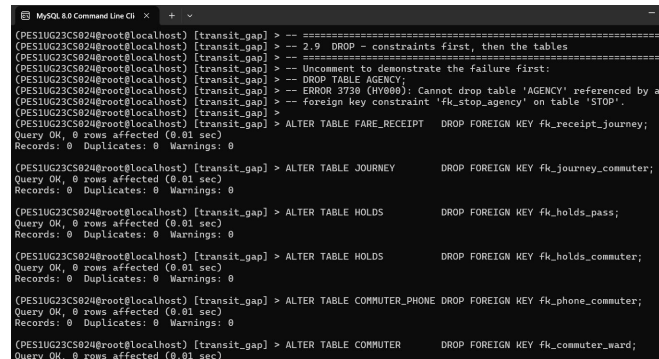
## 2.9 Drop the constraints, then drop the tables

```
ALTER TABLE FARE_RECEIPT DROP FOREIGN KEY fk_receipt_journey;
ALTER TABLE JOURNEY      DROP FOREIGN KEY fk_journey_commuter;
ALTER TABLE STOP         DROP FOREIGN KEY fk_stop_ward;
ALTER TABLE STOP         DROP FOREIGN KEY fk_stop_agency;
ALTER TABLE ROUTE        DROP FOREIGN KEY fk_route_trunk;
ALTER TABLE ROUTE        DROP FOREIGN KEY fk_route_agency;

DROP VIEW IF EXISTS COMMUTER_AGE, WARD_DENSITY;

DROP TABLE FARE_RECEIPT, JOURNEY, HOLDS, COMMUTER_PHONE, COMMUTER,
            PASS_TYPE, SERVES, PLATFORM, STOP, ROUTE, WARD, AGENCY;

SHOW TABLES;
```



```
MySQL 8.0 Command Line Client
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- 2.9 DROP - constraints first, then the tables
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- Uncomment to demonstrate the failure first:
(PES1UG23CS024@root@localhost) [transit_gap] > -- DROP TABLE AGENCY;
(PES1UG23CS024@root@localhost) [transit_gap] > -- ERROR 3730 (HY000): Cannot drop table 'AGENCY' referenced by a
(PES1UG23CS024@root@localhost) [transit_gap] > -- foreign key constraint 'fk_stop_agency' on table 'STOP'.
(PES1UG23CS024@root@localhost) [transit_gap] > --
(PES1UG23CS024@root@localhost) [transit_gap] > ALTER TABLE FARE_RECEIPT DROP FOREIGN KEY fk_receipt_journey;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] > ALTER TABLE JOURNEY DROP FOREIGN KEY fk_journey_commuter;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] > ALTER TABLE HOLDS DROP FOREIGN KEY fk_holds_pass;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] > ALTER TABLE HOLDS DROP FOREIGN KEY fk_holds_commuter;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] > ALTER TABLE COMMUTER_PHONE DROP FOREIGN KEY fk_phone_commuter;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] > ALTER TABLE COMMUTER DROP FOREIGN KEY fk_commuter_ward;
Query OK, 0 rows affected (0.01 sec)
```

### Write down the server's response first, then run it

1. STOP has PRIMARY KEY (Agency\_ID, Stop\_Code).  
What happens on a second insert of  
( 'BMTC','S001', ... )? And on ( 'BMRCL','S001', ... )?  
fk\_route\_trunk is ON DELETE SET NULL, and R501 and R502 both point at R500.
2. What do those two rows hold after R500 is deleted? What if the action were CASCADE?
3. ALTER TABLE PASS\_TYPE DROP COLUMN Pass\_Category; fails.  
What must run first, and why?
4. ALTER TABLE COMMUTER RENAME TO COMMUTERS;  
What does SHOW CREATE TABLE JOURNEY then report about its foreign key?
5. DROP TABLE AGENCY; is refused. Give the order of statements that makes it succeed.

**Every answer is visible in the output screenshots earlier in this deck**

1. Why is Stop\_Code alone not a valid primary key for STOP?  
A) it allows NULL    B) each agency issues its own codes, and they overlap  
C) it is a foreign key    D) it is computed from other columns
2. Which constraint on FARE\_RECEIPT makes one journey generate exactly one receipt?  
A) pk\_fare\_receipt    B) fk\_receipt\_journey    C) uq\_receipt\_journey    D) NOT NULL
3. Which referential action is declared on fk\_route\_trunk?  
A) ON DELETE CASCADE    B) ON DELETE SET NULL  
C) ON UPDATE CASCADE    D) NO ACTION
4. Age and Population\_Density are not stored as columns because  
A) MySQL forbids it    B) a stored copy goes stale    C) they are keys    D) of row size

### Quick check

1 - B      2 - C      3 - B      4 - B

### Predict the outcome

1. ERROR 1062 on the second ('BMTCL','S001'). ('BMRCL','S001') is accepted: a stop code is unique only within its agency.
2. R501 and R502 survive with Trunk\_Route\_ID set to NULL. Under CASCADE both feeders would be deleted with the trunk, and then their own feeders.
3. Drop chk\_pass\_category first - a column will not drop while a named constraint holds it.
4. InnoDB rewrites the reference, so JOURNEY now points at COMMUTERS.
5. Drop the foreign keys first, then drop the tables children before parents.



**THANK YOU**

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